

Task	Prompt	Answer
harmonic centrality	The task is to determine the harmonic centrality of a node in the graph. Harmonic centrality of a node u is the sum of the reciprocal of the shortest path distances from all other nodes to u. Here is a directed graph containing nodes from 1 to 8. The edges are: (6, 2), (6, 1), (6, 4), (6, 5), (6, 3), (7, 8). Question: What is the harmonic centrality of node 3 in the graph? You need to format your answer as a float number.	1.0
has cycle	The task is to determine if the graph has a cycle. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 2), (1, 4), (1, 5), (2, 4), (2, 5), (4, 9), (5, 3), (3, 6), (3, 8), (6, 8), (9, 7). Question: Does the graph have a cycle? Your answer should be Yes or No.	Yes
is bipartite	The task is to determine if the graph is bipartite. A bipartite graph is a graph whose nodes can be divided into two disjoint sets such that no two graph vertices within the same set are adjacent. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 4), (4, 3), (2, 5), (2, 3), (5, 6), (3, 6). Question: Is the graph bipartite? Your answer should be Yes or No.	Yes
is eularian	The task is to determine if the graph is Eulerian. An Eulerian graph is a graph that contains an Eulerian circuit, which is a cycle that visits every edge exactly once. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 5), (1, 3), (1, 2), (1, 4), (5, 2), (3, 2), (3, 4), (3, 6), (2, 4), (4, 6). Question: Is the graph Eulerian? Your answer should be Yes or No.	Yes
is regular	The task is to determine if the graph is regular. A regular graph is a graph where every node has the same degree. Here is an undirected graph containing nodes from 1 to 10. The edges are: (1, 5), (1, 7), (1, 10), (5, 2), (5, 10), (7, 8), (7, 10), (3, 9), (3, 8), (3, 4), (9, 4), (4, 6). Question: Is the graph regular? Your answer should be Yes or No.	No
is tournament	The task is to determine if the graph is a tournament. A tournament is a directed graph where every pair of nodes is connected by a single directed edge. The input graph is guaranteed to be directed. Here is a directed graph containing nodes from 1 to 10. The edges are: (1, 2), (2, 1), (2, 4), (4, 2), (4, 3), (3, 1), (5, 2), (5, 4), (6, 2), (6, 5), (7, 8), (8, 6), (9, 7), (10, 7). Question: Is the graph a tournament? Your answer should be Yes or No.	No